

Report

ALEKSEI KUZNETSOV

alexey.kuznetsov98@gmail.com

Abstract: In this work we experimentally demonstrate and analyze an interferometric 2×2 passive optical multiplier implemented on a silicon-on-insulator platform using the SiEPIC EBeam PDK. The circuit is composed of Mach–Zehnder interferometers (MZIs), broadband directional couplers (BDCs), multimode interferometers (MMIs), y-splitters and combiners. The topology is intended as a test structure and scalable building block for integrated optical matrix multipliers and photonic neural networks. We develop a compact matrix-based model that describes each passive element by a 2×2 complex transfer matrix with explicit phase, loss, and splitting-ratio errors, and use it together with Lumerical FDTD and INTERCONNECT simulations to predict circuit behavior. For simple MZI and MMI-based circuits, the measured spectral shapes and absolute power levels agree well with theory and simulation. For more complex topologies, including the full 2×2 multiplier, the average relative error between experiment and numerical model is about 21%, with an additional $\approx 13\%$ degradation when BDCs are used instead of MMIs in otherwise similar circuits. At the same time, the target interference patterns and suppression of unwanted modes are preserved. These results validate the proposed architecture and modeling methodology as a practical test platform for future active optical multipliers based on silicon photonics.

1. Introduction

Silicon photonics is considered one of the key approaches to implementing scalable, energy-efficient photonic integrated circuits for telecommunications, computing, and artificial intelligence systems. The high refractive index contrast of the silicon-on-insulator (SOI) platform, its compatibility with CMOS technology, and the availability of mature process design kits (PDKs) make it possible to realize complex passive and active components on a single chip, including waveguides, splitters, multimode interferometers, and modulators [1].

One of the most actively developing areas is photonic matrix multipliers and neuromorphic processors that use optical interferometric networks to implement matrix–vector operations. Various works have shown that meshes of Mach–Zehnder interferometers (MZIs) and other waveguide mesh architectures enable high-dimensional linear transformations and can accelerate computations compared with electronic implementations. At the same time, the basic building block of such circuits is a programmable 2×2 cell that represents a combination of splitters and phase shifters based on thermo-optic or carrier-based modulation [2–3].

For practical application of such architectures, it is necessary to achieve close agreement between modeling and experiment, as well as reproducible behavior of the passive elements that form the 2×2 cells. The characteristics of input/output grating couplers, Y-splitters, directional couplers (including broadband directional couplers, BDCs), and multimode interferometers (MMIs) are critical, since power distribution and interference circuits are built on top of them [4–5].

This work considers a non-resonant, interferometric approach to implementing a 2×2 multiplier based on waveguides, Y-splitters, directional couplers, and Mach–Zehnder interferometers on the SiEPIC platform. The main focus is on passive circuits implemented as test structures that make it possible to quantitatively compare numerical simulations with experimental measurements and to evaluate how different circuit elements contribute to the overall error [6].

In this work, the following structures are successively investigated (all on the SiEPIC platform):

- Mach–Zehnder interferometers (MZI) with different phase shifts between the arms.
- Broadband directional couplers (BDC) used as power splitters and combiners.
- Multimode interferometers (MMI) acting as compact splitters/combiners.
- Combined circuits that implement a passive 2×2 optical multiplier with fixed coefficients using combinations of MMIs, y splitters, BDCs and MMIs.

It is shown that for relatively simple structures such as a single MZI or an isolated splitter, the numerical models describe the experimental data with good accuracy, while the remaining discrepancies can be attributed to losses, technological variations, and imperfections of the PDK element models. As the complexity of the circuits increases, the sensitivity to phase errors and fabrication tolerances grows, which is reflected in higher deviations of the measured output powers from the model. Nevertheless, for the passive 2×2 multiplier realized within this work, the overall agreement between simulations and experiment is acceptable for teaching SiEPIC PDK and confirms the functional operability of the considered architectures.

2. Equipment and measurement methods

A custom-designed automated test setup [7–8] with automatic control software written in Python [9] was used to characterize the devices. An Agilent 81600B tunable laser served as the input source, and Agilent 81635A optical power sensors served as the output detectors. The wavelength was varied from 1500 to 1600 nm in 10 pm increments. Polarization-maintaining (PM) fiber [10] was used to maintain the polarization state of the light and to introduce TE polarization into the grating couplers. A 90° turn was used to couple light into the TM grating couplers [10]. A polarization-maintaining fiber array [11] was used to couple light into and out of the chip.

The standard relationship was used to convert the measured data into relative transmittance:

$$T = 10^{\frac{T_{dB}}{10}},$$

where T - normalized transmittance in relative units;

T_{dB} - original transmittance in dB obtained from the experiment.

The resulting dependences were further smoothed using a Savitzky–Golay filter (implemented in a Python library), which suppressed small-scale noise fluctuations and facilitated comparison with numerical simulations.

The relative error was calculated using the formula:

$$\varepsilon = \left| \frac{T_{exp} - T_{num}}{T_{num}} \right| \cdot 100\%.$$

It is worth noting that for small T_{num} , this metric significantly overestimates the error percentage; the true match quality is better described by the spectral shape and absolute differences.

To assess the impact of process variations on the circuit characteristics, statistical Monte Carlo simulation implemented in Python was used. Within the model, each basic element (Y-splitter, phase shifter, broadband directional coupler, and 2×2 multimode interferometer) was assigned a complex transmission matrix, accounting for losses and phase shifts. Then, in a 50,000-run cycle, a uniform random spread was added to the nominal values of the division ratios, phases, and losses, simulating process imbalance. Each run generated an output vector of complex amplitudes, from which the power at the outputs of interest was calculated, after which histograms of the output transmission distributions were constructed.

The basic idea is as follows. It is assumed that there is a vector of complex mode amplitudes in the waveguides, and each element of the FIS acts on this vector as a matrix. Then, when connecting the elements in series, we simply multiply the matrices and calculate the power as the square of the modulus of the desired vector component.

For example, there is an input complex field, which we can write as $\mathbf{E}_{in} = 1 + 0i$. A linear passive element is defined by a matrix $\mathbf{M}$ (usually unitary if there are no losses): $\mathbf{E}_{out} = \mathbf{M}\mathbf{E}_{in}$. If the elements are arranged sequentially along the propagation direction, with matrices $\mathbf{M}_1, \mathbf{M}_2, \dots, \mathbf{M}_k$, then for the entire cascade: $\mathbf{E}_{out} = \mathbf{M}_k \dots \mathbf{M}_2 \mathbf{M}_1 \mathbf{E}_{in}$. Then the final power can be obtained as $P = |\mathbf{E}_{out}|^2$.

For a 1×2 y splitter, which divides the input power into two waveguides and may have unsymmetrical division and additional phase shifts, it is convenient to use the following notation:

$$\mathbf{M}_y = \begin{pmatrix} \sqrt{\eta_1} e^{i\theta_1} \\ \sqrt{\eta_2} e^{i\theta_2} \end{pmatrix},$$

where $\eta_{1,2}$ - shares of power taking into account losses, $\eta_1 + \eta_2 \leq 1$;

$\theta_{1,2}$ - corresponding phases.

The inverse element of the Y-splitter, the Y-combiner (2×1), for matched operation is described by the Hermitian conjugate (transposed and complex-conjugated) matrix.

In general, a real MMI is described as follows. We assume that we have two input and two output waveguides, operate in only one propagation direction (from input to output), and do not model backward reflections. Then, the MMI is described by a 2×2 complex matrix:

$$\mathbf{M}_{MMI} = \begin{pmatrix} \sqrt{\eta_{11}} e^{i\theta_{11}} & \sqrt{\eta_{12}} e^{i\theta_{12}} \\ \sqrt{\eta_{21}} e^{i\theta_{21}} & \sqrt{\eta_{22}} e^{i\theta_{22}} \end{pmatrix},$$

where η_{mn} - shares of power when moving from the n -th input to the m -th output;

θ_{mn} - corresponding phases.

If the MMI is geometrically symmetrical vertically, both inputs are equivalent and the MMI is reciprocal, then the conditions are satisfied $|\sqrt{\eta_{11}} e^{i\theta_{11}}| = |\sqrt{\eta_{22}} e^{i\theta_{22}}|$, $|\sqrt{\eta_{12}} e^{i\theta_{12}}| = |\sqrt{\eta_{21}} e^{i\theta_{21}}|$.

In the case where two waveguides (two arms of the MZI) are considered and the phase changes in both arms, it is convenient to introduce a diagonal 2×2 matrix to describe such a structure:

$$\mathbf{M}_{MZI} = \begin{pmatrix} \sqrt{\eta_{11}} e^{i\theta_{11}} & 0 \\ 0 & \sqrt{\eta_{22}} e^{i\theta_{22}} \end{pmatrix}.$$

In the idealized lossless case, the matrices are unitary, and real losses are introduced by reducing the moduli of the matrix elements. This matrix formalism allows for direct comparison of simulation results in Python, Lumerical INTERCONNECT, and experimental data.

For full-wave simulation, the Lumerical FDTD software package was used. It makes it possible to obtain mode field distributions, automatically calculate the necessary waveguide parameters (such as the effective refractive index), and carry out parametric studies to better understand system tolerances.

3. Waveguide characteristics

For full-wave simulations, the Lumerical FDTD package was used. It allows one to obtain mode field distributions, automatically calculate the required waveguide parameters such as effective refractive index, and perform parametric studies for a better understanding of system tolerances. At the first stage, a straight strip SOI waveguide with width $w = 500$ nm and height $h = 220$ nm was characterized. Using a mode solver in the C band around 1550 nm, the normalized electric field profile of the fundamental TE-like mode was obtained (Figure 1), and the indices at wavelength 1550 nm were extracted: $n_{\text{eff}} = 2.354545$ и $n_g = 4.518960$.

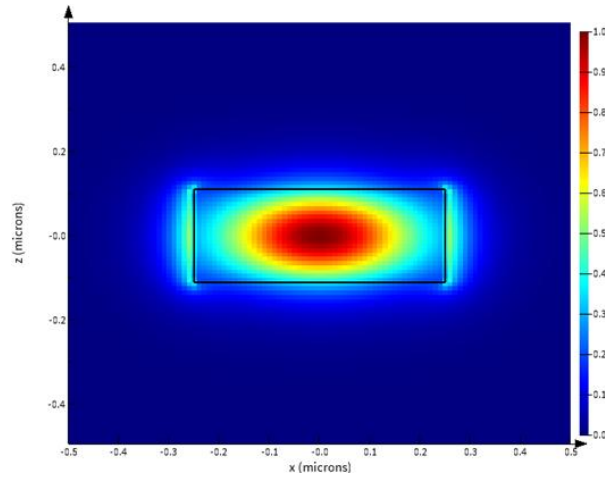


Fig. 1. Distribution of the electric field norm of the fundamental mode of a waveguide in a transverse section. Color denotes the field in relative units, where red is the maximum and blue is the minimum.

Then a wavelength sweep was performed to obtain the spectral dependences $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$ in the target range. The calculated mode profile confirms well confined single mode guiding in the C band for the chosen cross section, and the obtained $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$ trends shown in Figure 2 are later used as input data for analyzing the MZI.

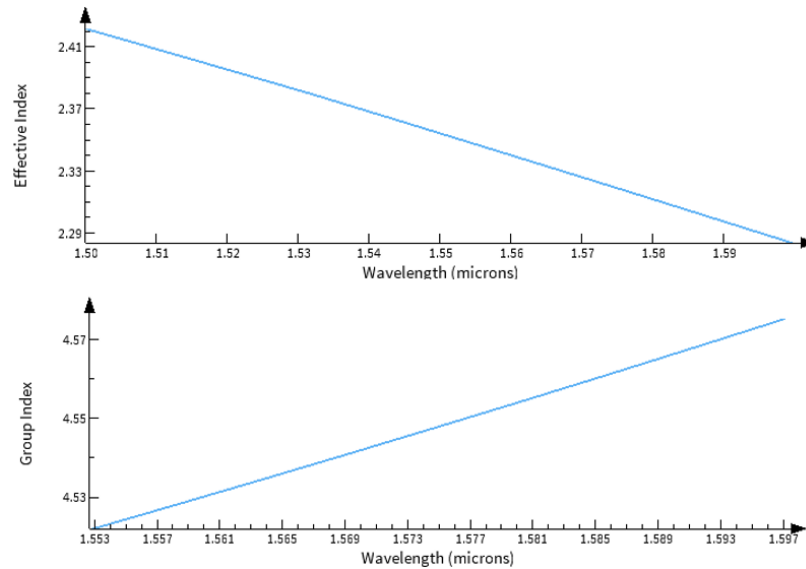


Fig. 2. Graphs of the dependence of the effective refractive index and group index of the waveguide on the wavelength.

4. Test structure

To verify the modeling methodology and the processing of experimental data, a test structure of the form “grating coupler → waveguide → grating coupler” was used. The layout of the structure was designed in KLayout, and numerical simulations were carried out in Lumerical INTERCONNECT using the built-in S-parameters provided by the SiEPIC foundry.

To compare numerical and experimental results, the spectra of all subsequent circuits were normalized to the transmission spectrum of this test structure. Experimental data given in decibels were converted to relative transmission, smoothed with a Savitzky–Golay filter, and divided by the measured spectrum of the “grating coupler → waveguide → grating coupler” structure. The numerical results were normalized in the same way using the calculated transmission of the same reference system. This normalization approach makes it possible to simultaneously account for the real input/output losses through the grating couplers and for a possible technological shift of the grating coupler transmission maximum relative to the nominal PDK wavelength. In what follows, the terms “test channel” or “reference channel” denote precisely this structure, which is used as the baseline for all subsequent “simulation vs experiment” comparisons.

The general view of the test structure in KLayout is shown in Figure 3.

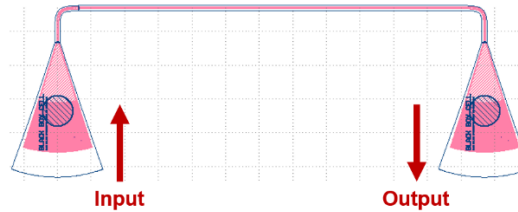


Fig. 3. General view of the test circuit in KLayout, consisting of grating coupler → waveguide → grating coupler.

Further on, Figure 4 shows a comparison of the transmission spectra calculated using numerical methods and measured in the experiment: the top row shows the data in decibels, the bottom row shows the same data in relative units; the left column corresponds to calculations in Lumerical INTERCONNECT, the right to real measurements.

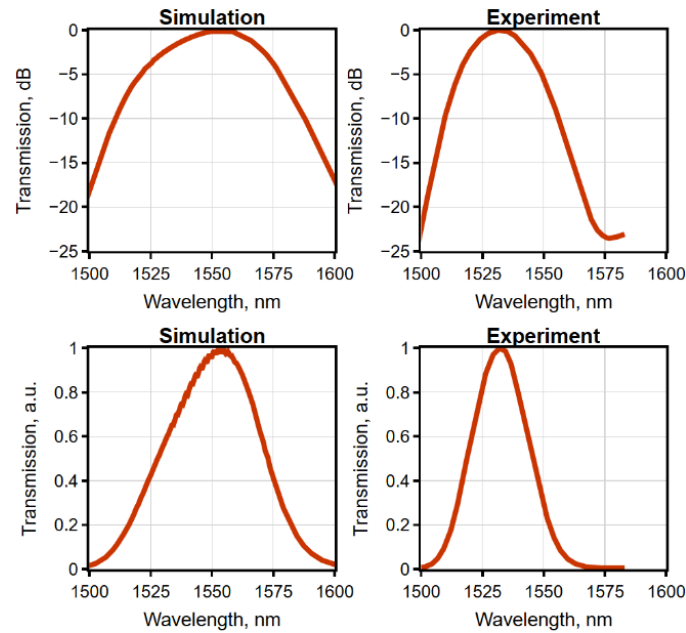


Fig. 4. Graphs of the transmission dependence of the grating coupler → waveguide → grating coupler test circuit. The top row is in dB, the bottom row is in relative units. The left column shows calculations performed in Lumerical INTERCONNECT, the right column shows experimental measurements.

The numerical and experimental spectra have similar shapes and exhibit a characteristic bandpass dependence determined by the spectral response of the grating couplers. In subsequent sections, the main circuits were normalized to the measured spectrum of the test structure, which allowed us to account for actual input/output losses and the technological shift of the maximum transmission relative to the calculated values. Thus, the test structure was designated as the reference channel for subsequent data normalization.

5. Mach–Zehnder interferometers

A Mach–Zehnder interferometer (MZI) was used as the basic element for the study. It generates specified transmittances at the outputs by means of a controlled phase shift between its arms. The phase shift is achieved by adding an additional section of length Δl to one of the arms, after which interference occurs in the output Y-splitter. Three characteristic modes were chosen for the test circuit, given the limited number of available outputs:

- **Constructive interference.** The output transmittance is close to unity. The phase shift between the arms is $\Delta\phi = 0 \pm 2\pi n$, $n \in \mathbb{Z}$. This configuration is implemented along a trajectory from Input to Output 1.
 - **Intermediate interference.** The output transmittance is close to 0.5, corresponding to a phase shift of $\Delta\phi = \pi/2 \pm \pi n$, where n is an integer. The required elongation of one of the arms was determined based on Lumerical INTERCONNECT simulation results and amounted to $\Delta l = 0.1585 \mu\text{m}$. This configuration is implemented along a trajectory from Input to Output 2.
 - **Destructive interference.** At the output, we obtain transmission tending to zero. The phase shift between the arms is $\Delta\phi = \pi \pm 2\pi n$, where n is an integer. The "increment" in the arm was empirically determined in Lumerical INTERCONNECT to achieve the required phase shift of $\Delta l = 0.317 \mu\text{m}$. This configuration is implemented along a trajectory from Input to Output 3.
- These interferometer variants were implemented in the KLayout design environment. The geometry of the corresponding structures is hereinafter referred to as **Circuit 1** and is shown in Figure 5.

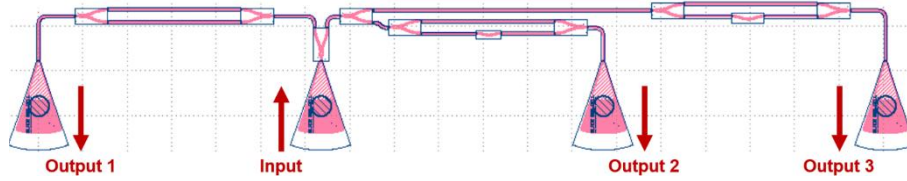


Fig. 5. General view of Circuit 1 in KLayout, consisting of grating couplers, y-splitters, waveguides and Mach-Zehnder interferometers.

Assuming an input power of unity and an ideal Y-splitter, the first Y-splitter divides the power equally between the arms. On the way to Output 1, constructive interference at the output yields an effective transmittance of 0.5. The remaining 0.5 power is again divided by the second Y-splitter, so that each subsequent module receives 0.25. At Output 2, in the intermediate interference mode, the power is divided in half, yielding a transmittance of 0.125, while at Output 3, with destructive interference, a value close to zero is expected.

The described circuit was also modeled in Lumerical INTERCONNECT; the system's appearance is shown in Figure 6.

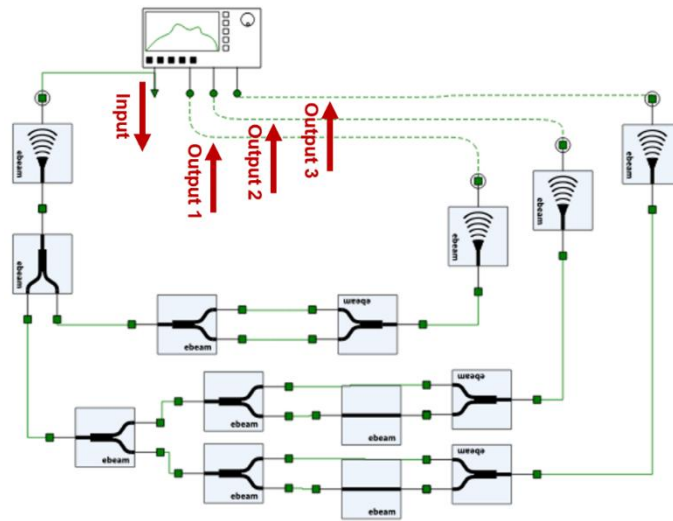


Fig. 6. Circuit 1, implemented in Lumerical INTERCONNECT, consisting of an optical network analyzer, grating couplers, Y-splitters, and waveguides.

Figure 7 below shows the theoretical expectations, numerical results, and experimental data for Circuit 1. The "Simulation" column shows the numerical simulation results in Lumerical INTERCONNECT, where the optical parameters for each element are used. This allows realistic simulation of the circuit's behavior before fabrication. The "Experiment" column contains data obtained from measurements of an experimental sample fabricated at the SiEPIC fab.

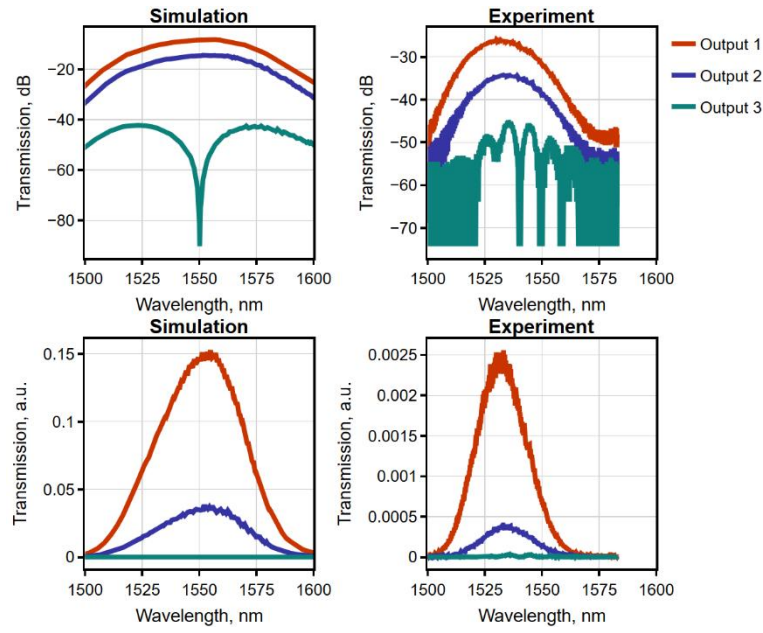


Fig. 7. Initial data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The obtained results were processed according to the approach described in the methodology and normalized to the spectra of the test circuit; the results are presented in Figure 8.

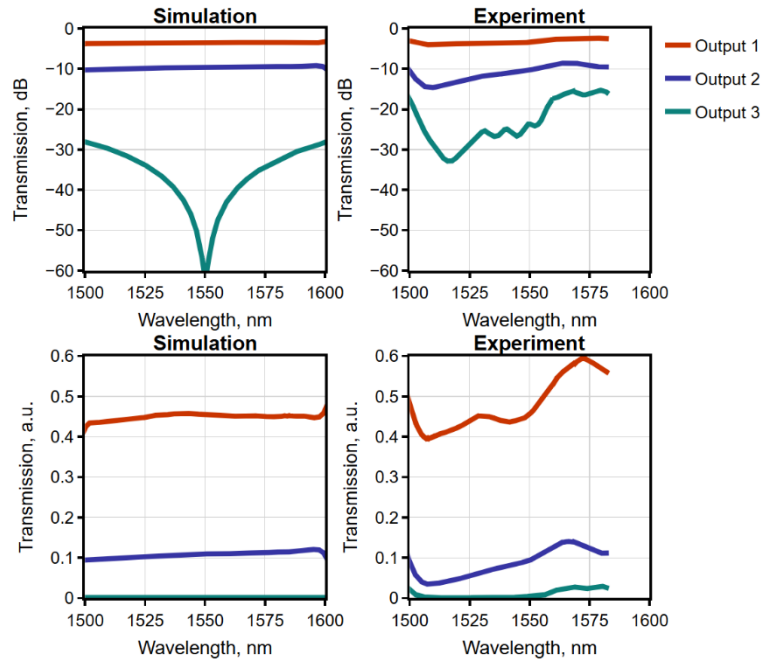


Fig. 8. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

A comparison of theoretical and experimental data was performed using transmission in relative units. The unevenness of the experimental curve is due to the normalization spectrum peaking at approximately 1530 nm, which differs from the calculated value due to technological variation in the grating coupler parameters. Since the transmission spectrum is close to Gaussian in shape, normalization to it introduces numerical errors at the edges of the range due to division by small values, which amplifies any irregularities in the original experimental data.

To understand possible deviations from the ideal expected result, a statistical analysis of the system was performed using the Monte Carlo method. For each element, approximate parameter values and their technological variation were selected so that the experimental data fell within the numerical range, providing an approximate estimate of the technological errors of real devices. For the Y-splitter, the loss was 0.26 dB, the phase shift errors at each output were $\pm 3^\circ$, and the error in the ideality of the 50:50 splitting ratio was 5%. The phase adjustment tolerance in the modulator

is $\pm 3^\circ$. It is worth noting that these values are empirically determined, so in reality, other combinations may occur that result in similar distribution values. To truly understand manufacturing inaccuracies, scanning electron microscope images and additional phase analysis are required. The power distribution graphs are shown in Figure 9.

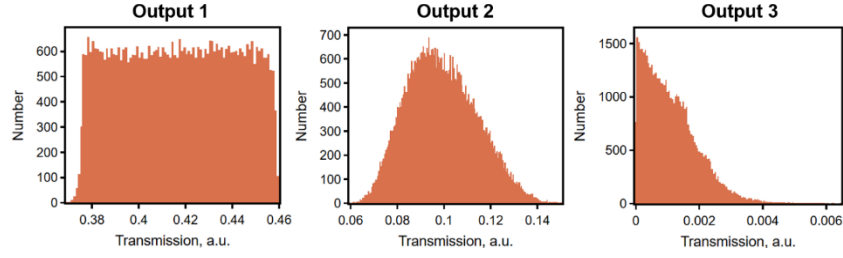


Fig. 9. Probability of power distribution at each of the outputs of Circuit 1.

For output 1, corresponding to the $\Delta\phi \approx 0$ mode, the resulting power distribution was close to uniform. In the standard MZI model, the output port transfer is described by an expression of the form:

$$T(\Delta\phi) = T_0 + T_1 \cos(\Delta\phi + \phi_0).$$

That is, the interference term is present at all values of $\Delta\phi$ and reaches its maximum contribution precisely at $\Delta\phi = 0$. However, in the vicinity of $\Delta\phi = 0$, the phase derivative of T is small, so small phase noise of the order of $\pm 3^\circ$ has little effect on transmission, and the power spread is determined primarily by variations in the division factors and losses. In the adopted Monte Carlo model, this leads to a nearly uniform distribution. For $\Delta\phi \approx \pi/2$ and π , the dependence of transmission on phase is steeper, and even a small phase spread leads to a quasi-normal power distribution.

In addition, the effective refractive index of the waveguide can be estimated experimentally using the well-known formula:

$$\Delta\phi = \frac{2\pi}{\lambda} n_{eff} \Delta l \Rightarrow n_{eff} = \frac{\lambda \Delta\phi}{2\pi \Delta l}.$$

For the case under consideration, where $\Delta\phi = \pi$, $\Delta l = 0.317 \mu\text{m}$, and $\lambda = 1550 \text{ nm}$, this yields $n_{eff} = 2.4448$. This value is several percent higher than $n_{eff} = 2.35$, calculated by the mode solver for a straight section of waveguide. This discrepancy is expected, since the effective optical length of the interferometer can differ from the purely geometric elongation Δl due to bends, transitions, and mode index inhomogeneities. Furthermore, a slight temperature drift could have occurred during the measurement. Taken together, this yields a reasonable estimate of n_{eff} for the waveguide under consideration.

Since the theoretical model (which estimated the coefficients at the output of an ideal circuit) did not take into account element losses, it is advisable to estimate the errors by comparing the experiment with the numerical simulation. Summary values at a wavelength of 1550 nm are given in Table 1.

Table 1. Output transmission values obtained for circuit 1.

		Output 1	Output 2	Output 3
Circuit 1	Theory	0.5	0.125	0
	Simulation	0.46	0.11	0
	Experiment	0.45	0.1	0
	Error	2.2%	9.1%	0%

The numerical and experimental data deviate from the theoretical values in one direction and by a comparable amount, indicating systematic losses and manufacturing tolerances rather than a fundamental error in the model. The obtained relative errors, on the order of a few percent, are within a reasonable range for modern silicon photonic technologies.

6. Splitters

6.1 Circuit with broadband directional couplers

To analyze the behavior of a broadband directional coupler (BDC) in Lumerical INTERCONNECT, a test circuit consisting of a source, a Y-splitter, a BDC, and a receiver was assembled. Unlike a classic 3 dB directional coupler, which evenly divides power between two outputs, the BDC from the SiEPIC library provides an asymmetric power distribution at the outputs and exploits waveguide asymmetry to extend the bandwidth [4].

The circuit diagrams and simulation results are shown in Figure 10. For convenience, the coupler inputs and outputs are numbered according to the PDK convention. The vertical axis shows the circuit schematic and the corresponding numerical simulation results.

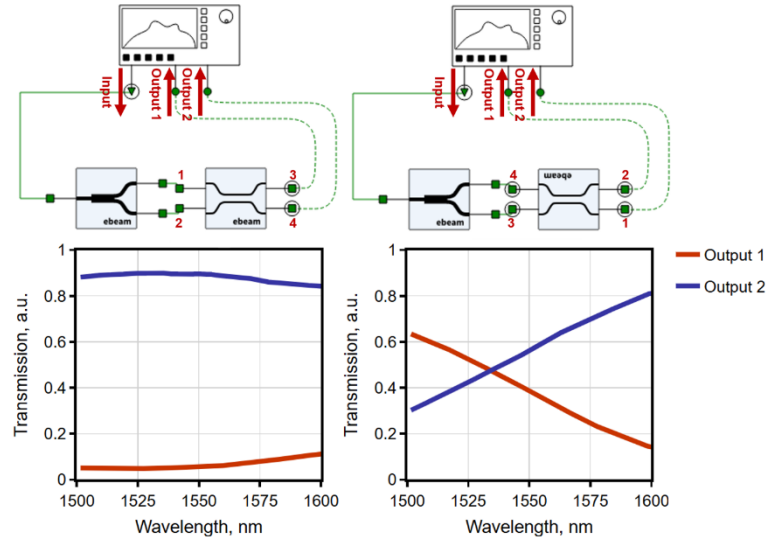


Fig. 10. Top row: test circuits implemented in Lumerical INTERCONNECT, consisting of an optical network analyzer, a y-splitter, and a broadband directional coupler. Bottom row: data obtained numerically from the presented circuits, transmission is given in relative units. Color indicates correspondence to specific outputs on the circuit.

According to the numerical simulations, with the BDC connected in the forward direction at a wavelength of 1550 nm, the powers at outputs 4 and 3 are approximately 0.9 and 0.06, respectively, while with the BDC rotated by 180° (reverse connection) the powers at outputs 1 and 2 are approximately 0.56 and 0.39.

Next, the combination of these output signals was numerically studied using a Y-splitter. The circuit and simulation results are shown in Figure 11.

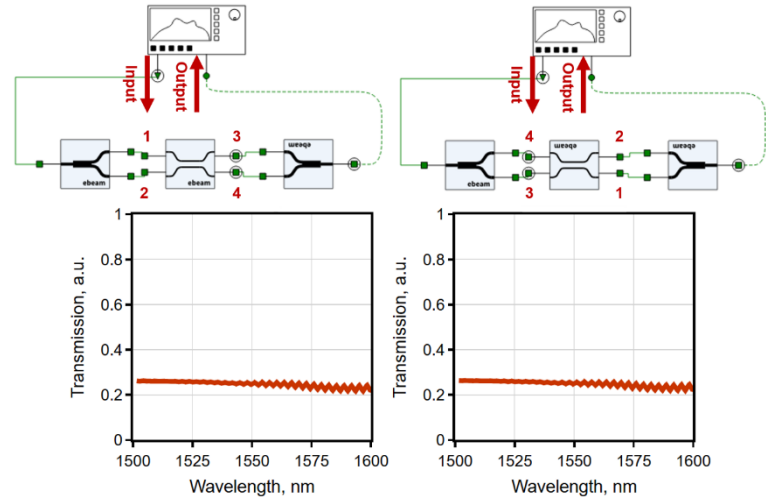


Fig. 11. Top row: test circuits implemented in Lumerical INTERCONNECT, consisting of an Optical Network Analyzer, y-splitters, and a broadband directional coupler. Bottom row: data obtained numerically from the presented circuits, transmission is given in relative units.

By combining the BDC outputs with a Y-splitter, a transmittance close to 0.25 is obtained at the selected output of the resulting circuit. Rotating the BDC by 180° in this configuration has virtually no effect on the final power distribution.

These interferometer variants, combined with Y-splitters and BDCs, were implemented in the KLayout design environment. The geometry of the corresponding structures is referred to below as **Circuit 2** and is shown in Figure 12.

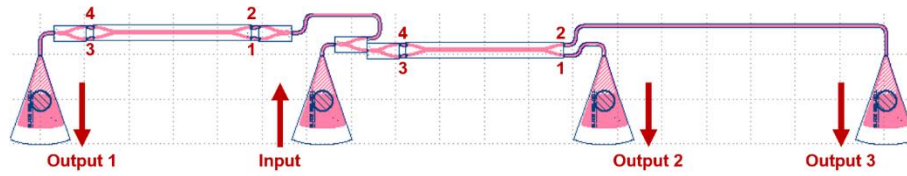


Fig. 12. General view of Circuit 2 in KLayout, consisting of grating couplers, y-splitters, waveguides, Mach-Zehnder interferometers and broadband directional couplers.

For an ideal, lossless case, the expected output powers can be estimated as follows. Assume the input power is unity. The first Y-splitter divides the power equally between the arms: 0.5 in the left and right arms. The left arm contains two Mach-Zehnder interferometers with identical arms; in this configuration, Output 1 receives a transmittance of 0.125. The right arm carries a power of 0.5. Given the previously obtained characteristics of the selected BDC orientation, this power is distributed between Outputs 2 and 3 as 0.2 and 0.28, respectively.

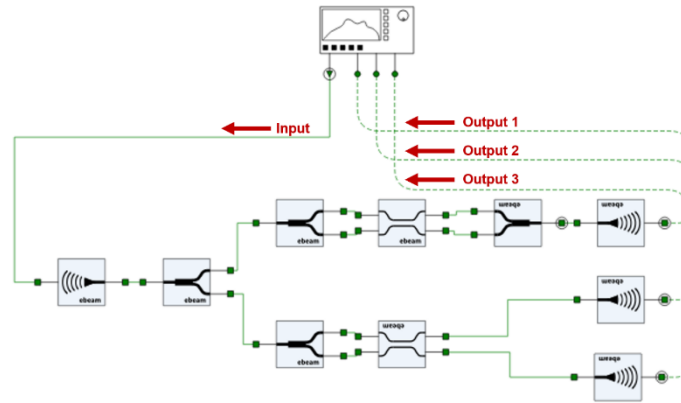


Fig. 13. Circuit 2, implemented in Lumerical INTERCONNECT, consisting of Optical network analyzer, grating couplers, y-splitters, waveguides and broadband directional couplers.

Figure 14 below shows the combined numerical results and experimental data for Circuit 2. The "Simulation" column displays the numerical simulation results in Lumerical INTERCONNECT, where the optical parameters for each element are used. This allows for realistic simulation of the circuit's behavior before fabrication. The "Experiment" column contains data obtained from measurements of an experimental sample fabricated at the SiEPIC fab.

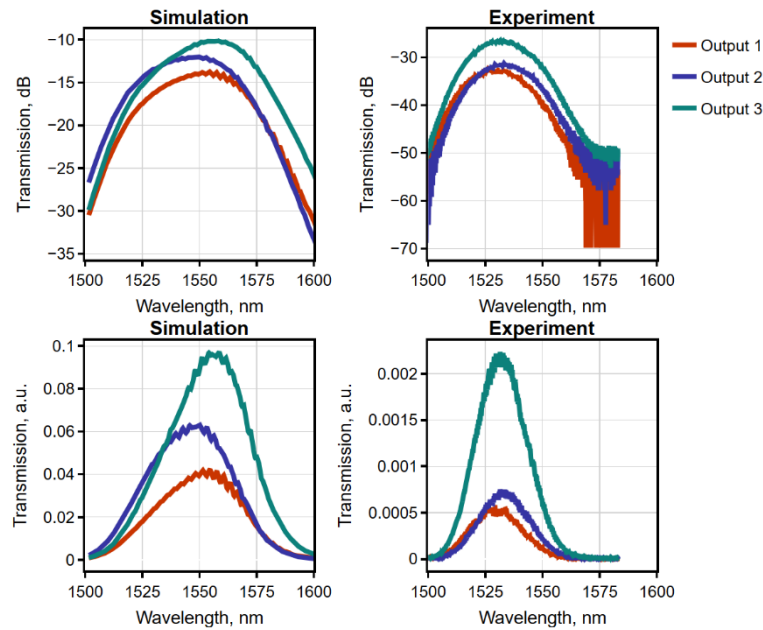


Fig. 14. Initial data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The experimental and numerical data were processed using the same normalization and smoothing procedure as described previously; the results are presented in Figure 15.

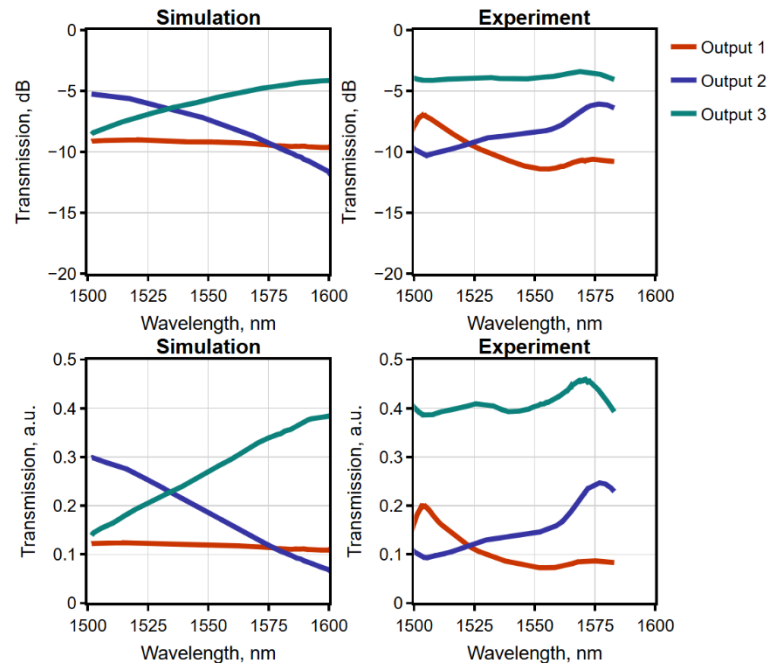


Fig. 15. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

Summary values at 1550 nm are given in Table 2.

Table 2. Output transmission values obtained for circuit 2.

		Output 1	Output 2	Output 3
Circuit 2	Theory	0.125	0.2	0.28
	Simulation	0.12	0.18	0.27
	Experiment	0.07	0.14	0.4
	Error	42%	22%	48%

Although the experimental spectral shape retains the characteristic interference pattern, significant quantitative discrepancies indicate high sensitivity to phase errors, BDC geometric tolerances, and additional losses in the fabricated sample. This is consistent with the fact that broadband directional couplers in silicon typically impose more stringent process accuracy requirements than classic 3 dB DC and MMI couplers. It should also be remembered that for small T_{num} the error metric significantly overestimates the percentage error; in some cases the true match quality is better described by the spectral shape and absolute deviations.

6.2 Circuit with multimode interferometers

This section analyzes a circuit in which broadband directional couplers (BDC) are replaced with 2×2 multimode interferometers (MMI). In an ideal, lossless case with a Y-splitter that divides the power equally between the arms and a subsequent signal split in one arm by the MMI, the expected transmittances at the three outputs are 0.5, 0.25, and 0.25.

Since the MMI element in the PDK used is still in a test stage and its S-parameters are not integrated into the library, for simulation the MMI was approximated by a directional coupler (DC) with matched amplitudes and phases.

The circuit combining grating couplers, Y-splitters, the MMI, and waveguides was implemented in the KLayout design environment. The geometry of the corresponding structures is hereinafter referred to as Circuit 3 and is shown in Figure 16.

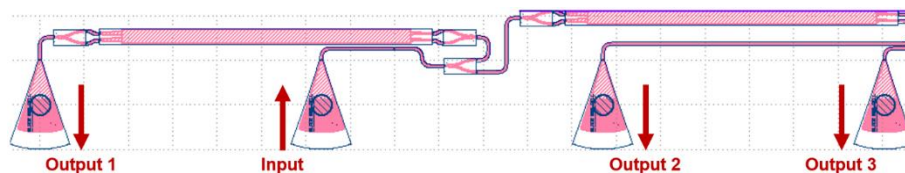


Fig. 16. General view of Circuit 3 in KLayout, consisting of grating couplers, y-splitters, waveguides, Mach-Zehnder interferometers and multimode interferometers.

Figure 17 shows the numerical and experimental results for **Circuit 3**.

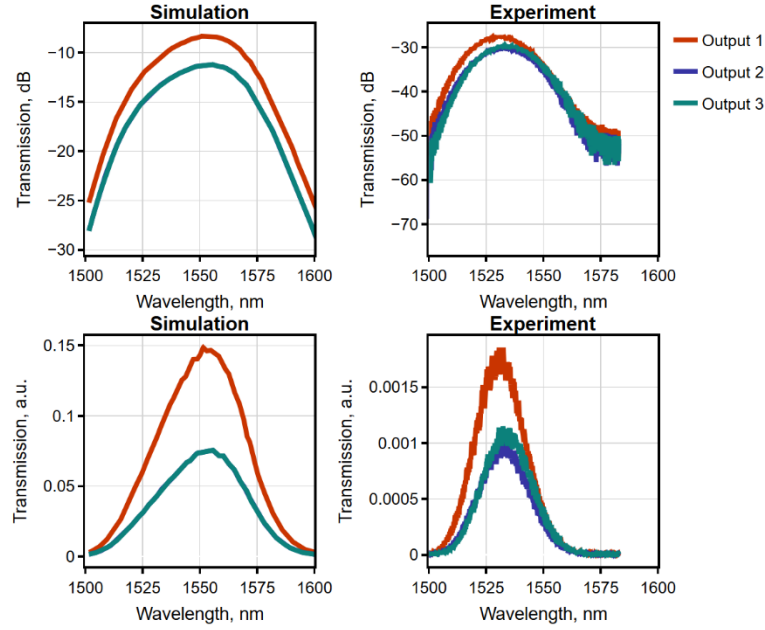


Fig. 17. Initial data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The original experimental and numerical data were transformed similarly to the methods discussed previously. The results are presented in Figure 18.

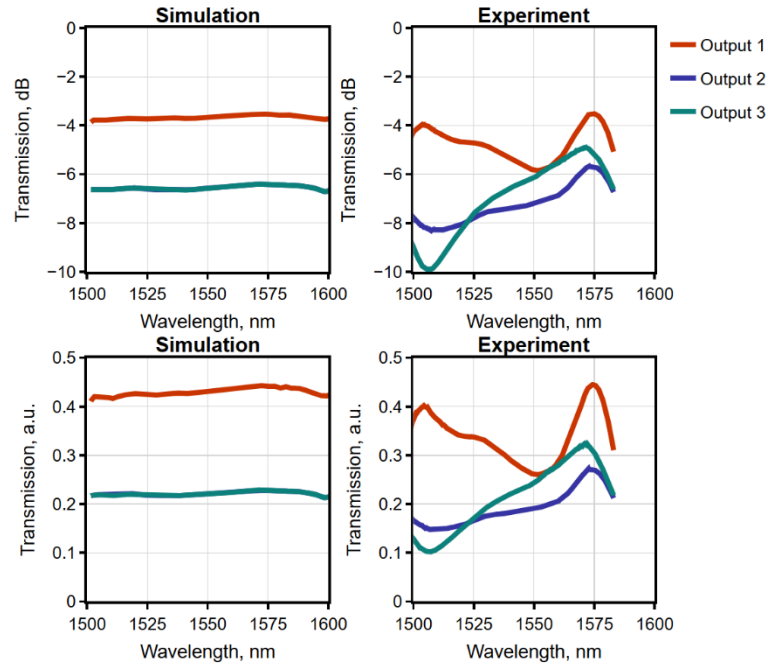


Fig. 18. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The resulting circuit also underwent a statistical error analysis as described previously. For the MMI, the element loss was assumed to be 0.3 dB, and the phase and power errors at the split were taken to be 10%. The resulting distributions are shown in Figure 19.

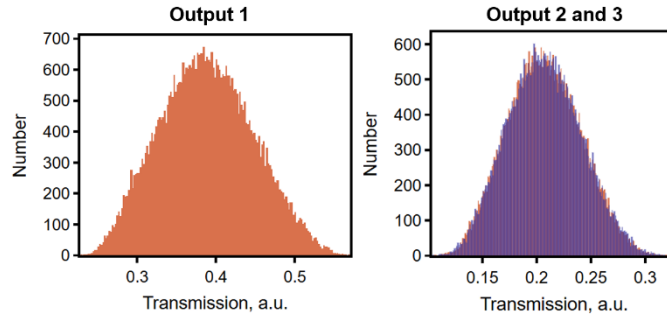


Fig. 19. Probability of power distribution at each of the outputs of Circuit 3.

The MMI error values are slightly overestimated to match the experimentally obtained values; in reality, such deviations may be due to fabrication inaccuracies or measurement errors.

Summary values at a wavelength of 1550 nm are presented in Table 3.

Table 3. Output transmission values obtained for circuit 3.

		Output 1	Output 2	Output 3
Circuit 3	Theory	0.5	0.25	0.25
	Simulation	0.43	0.22	0.22
	Experiment	0.26	0.19	0.25
	Error	40%	14%	14%

Overall, the MMI based circuit qualitatively reproduces the required power distribution, and quantitative deviations for most channels are smaller than in the BDC version. The largest difference is observed at Output 1, which may be due to systematic losses and the inaccuracy of the MMI approximation via DC. In general, the MMI exhibits more predictable behavior and better tolerance to variations compared to the BDC, which is consistent with the general notion of MMIs as technology tolerant components.

7. Combination of splitters and Mach-Zehnder interferometers

7.1 Circuit with broadband directional couplers

Next, we analyzed a circuit combining a Mach-Zehnder interferometer with a phase shift of $\pi/2$ in one arm due to the addition of an extension of $\Delta l = 0.1585 \mu\text{m}$ and a BDC. As in the previous circuits, we first simulated a simple circuit consisting of a source, Y-splitter, arm extension, BDC, and receiver. For convenience, the coupler inputs and outputs are numbered as in the PDK. The circuit and the corresponding numerical simulation results are shown in Figure 20.

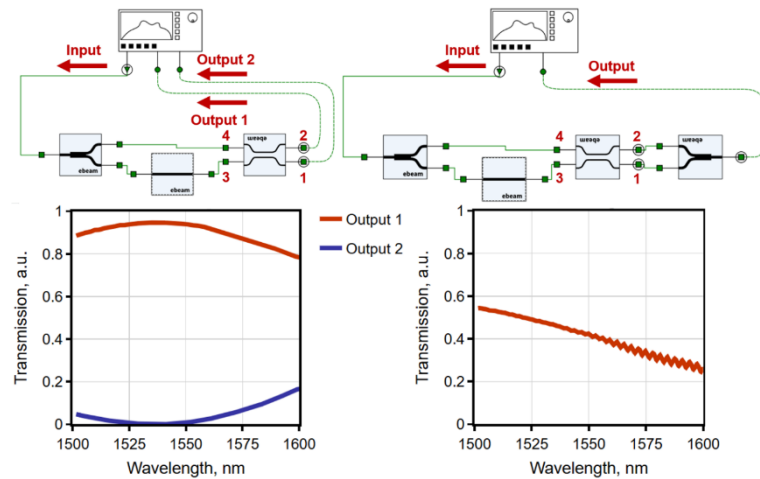


Fig. 20. Top row: Test circuits implemented in Lumerical INTERCONNECT, consisting of an Optical Network Analyzer, a Y-splitter, and broadband directional couplers. Bottom row: Data obtained numerically from the presented circuits, transmission is given in relative units. Color indicates correspondence to specific outputs on the circuit.

Based on the simulation data, the following transmittance values can be obtained for each of the BDC outputs at a wavelength of 1550 nm. With the BDC connected in reverse, the transmittance at output 2 is approximately 0.94 and at output 1 approximately 0.01. If, after this circuit, a Y-splitter is connected and both BDC outputs are fed to it,

then, taking into account the phase difference between the signals, the numerical simulation shows that the selected Y-splitter output at 1550 nm yields a transmittance of approximately 0.42.

These interferometer variants, combined with Y-splitters and BDCs, were implemented in KLayout. The geometry of the corresponding structures is hereinafter referred to as Circuit 4 and is shown in Figure 21.

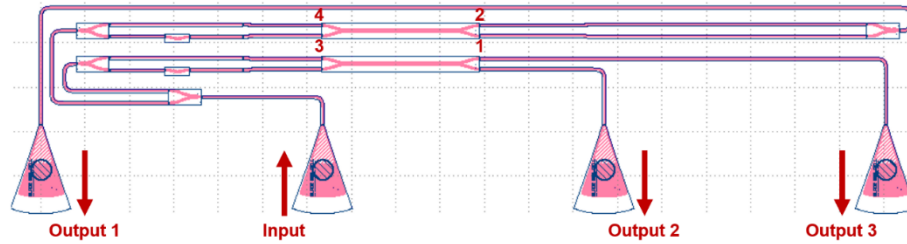


Fig. 21. General view of Circuit 4 in KLayout, consisting of grating couplers, y-splitters, waveguides, Mach-Zehnder interferometers and broadband directional couplers.

In the idealized lossless case, using the Mach-Zehnder interferometer with a $\pi/2$ phase shift and the chosen BDC orientation, the expected transmittances at the three analyzed outputs of Circuit 4 are 0.25, 0, and 0.5. These values were obtained from a model in which the Y-splitter divides the power equally between the arms and the BDC redistributes the power among the outputs in accordance with the previously measured splitting ratios. To verify the described circuit, numerical simulations were also performed in Lumerical INTERCONNECT; the system diagram is shown in Figure 22.

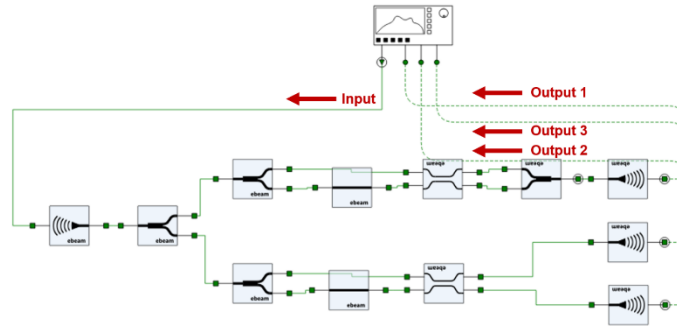


Fig. 22. Circuit 4, implemented in Lumerical INTERCONNECT, consisting of Optical network analyzer, grating couplers, y-splitters, waveguides and Broadband directional coupler.

Figure 23 shows the numerical and experimental results for Circuit 4.

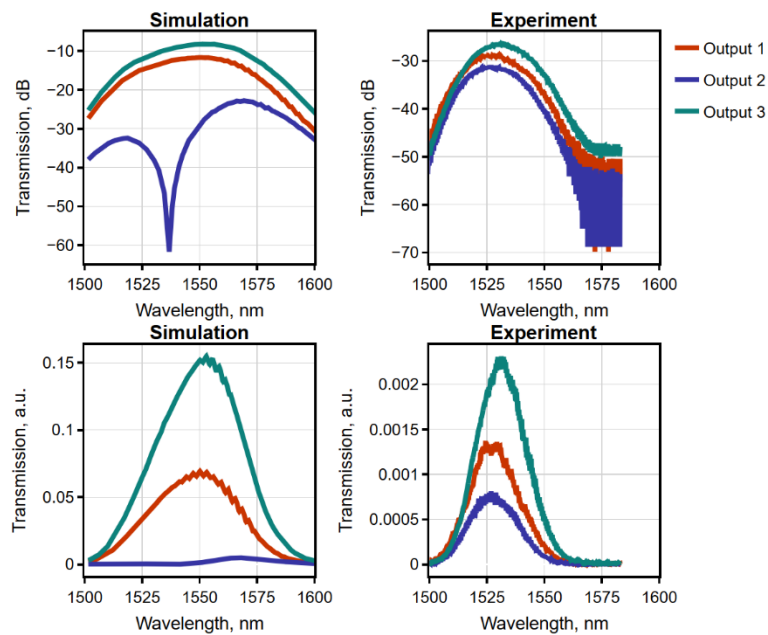


Fig. 23. Initial data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

Figure 24 shows the normalized and smoothed data.

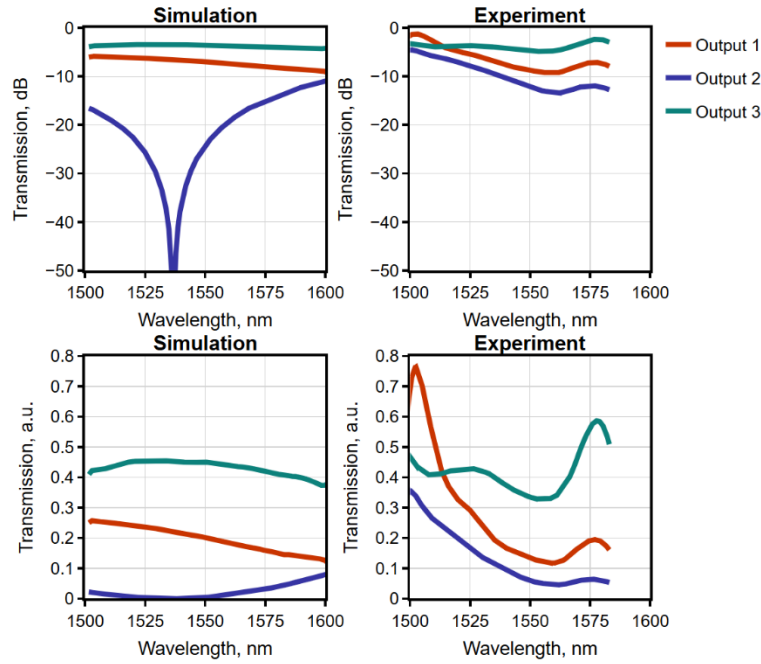


Fig. 24. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The summary values at a wavelength of 1550 nm are given in Table 4. It is worth noting that for outputs with ideal zero, the error was not calculated using a formula but is interpreted qualitatively.

Table 4. Output transmission values obtained for circuit 4.

		Output 1	Output 2	Output 3
Circuit 4	Theory	0.25	0	0.5
	Simulation	0.20	0	0.45
	Experiment	0.13	0.06	0.33
	Error	35%	0%	27%

In this section, we investigated a circuit combining a Mach–Zehnder interferometer with a $\pi/2$ phase shift in one arm and a broadband directional coupler. Taken together, the results for Circuit 4 demonstrate that the combination of Y-splitters, Mach–Zehnder interferometers, and broadband directional couplers allows for flexible power distribution between outputs and the formation of channels with suppressed signals but requires precise control of phase relationships and BDC parameters to achieve quantitative agreement with calculations.

7.2 Circuit with multimode interferometers

As noted earlier, in Lumerical INTERCONNECT the multimode interferometer was modeled by a directional coupler with the same amplitudes and phases. For the combination of the MMI with the MZI that introduces a phase shift of $\pi/2$ via the additional elongation $\Delta l = 0.1585 \mu\text{m}$, the simulated values are of the same order of magnitude as those for the BDC with the same MZI.

According to the simulations, at a wavelength of 1550 nm the transmittance at MMI output 2 is approximately 0.93 and at output 1 approximately 0. If these two MMI outputs are then connected to a Y-splitter, taking into account the phase relationship between the signals, the selected Y-splitter output yields a transmittance of approximately 0.45 at 1550 nm.

The interferometer designs described above, combined with Y-splitters and MMIs, were implemented in KLayout. The geometry of the corresponding structures is referred to below as **Circuit 5** and is shown in Figure 25.

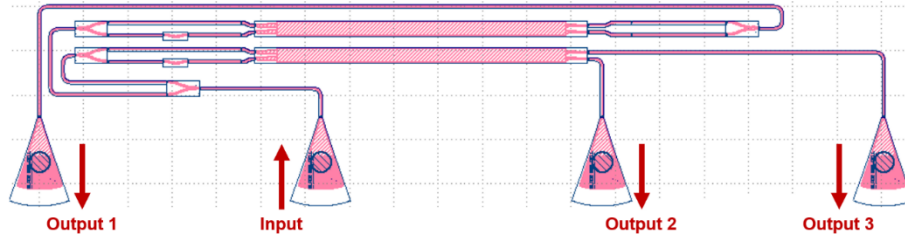


Fig. 25. General view of Circuit 5 in KLayout, consisting of grating couplers, y-splitters, waveguides, Mach-Zehnder interferometers and multimode interferometers.

In the idealized case of lossless operation, using the described Mach-Zehnder interferometer with a $\pi/2$ phase shift and MMI, the expected transmittances at the three analyzed outputs of Circuit 5 are 0.25, 0, and 0.5. These values were obtained from a model in which the y-splitter divides the power equally between the arms, and the MMI redistributes it among the outputs according to the previously measured division ratios.

Figure 26 below shows the combined numerical results and experimental data for Circuit 5. The "Simulation" column shows the numerical simulation results in Lumerical INTERCONNECT, where the optical parameters of each element represented in the PDK library are used, and the multimode interferometer is approximated by a directional coupler matched for amplitudes and phases.

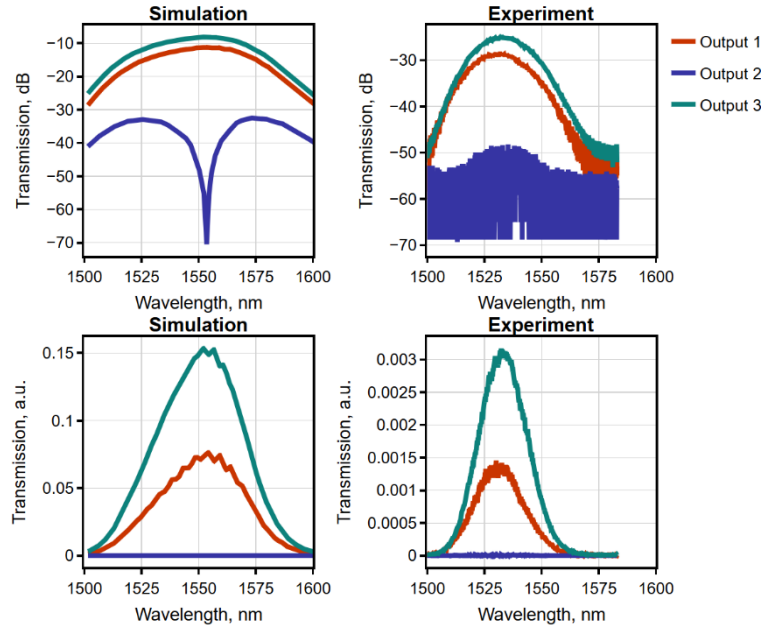


Fig. 26. Initial data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The original experimental and numerical data were transformed similarly to earlier methods. The results are presented in Figure 27.

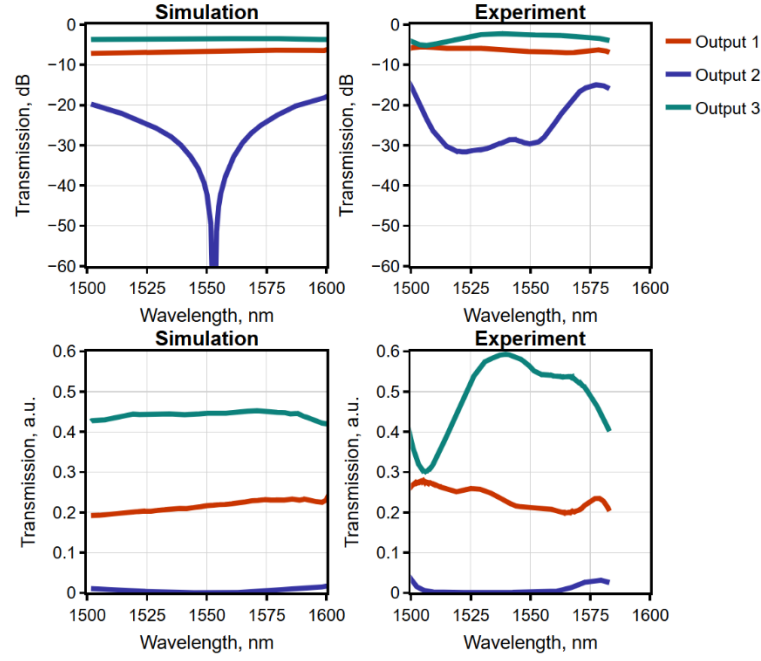


Fig. 27. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

The summary values at a wavelength of 1550 nm are given in Table 5. It is worth noting that for outputs with ideal zero, the error was not calculated using a formula, but is interpreted qualitatively.

Table 5. Output transmission values obtained for circuit 5.

		Output 1	Output 2	Output 3
Circuit 5	Theory	0.25	0	0.5
	Simulation	0.22	0	0.45
	Experiment	0.21	0	0.56
	Error	5%	0%	24%

In this section, we investigated a circuit combining a Mach–Zehnder interferometer with a $\pi/2$ phase shift in one arm and a 2×2 MMI multimode interferometer. Taken together, the results demonstrate that the combination of Y-splitters, Mach–Zehnder interferometers, and multimode interferometers enables the desired power distribution between outputs and the formation of a signal suppressed channel, while the circuit behavior is more intuitively predictable compared to the use of broadband directional couplers. Further calibration of the MMI models and consideration of fabrication tolerances are required to achieve precise quantitative agreement with experiment.

8. 2x2 passive optical multiplier

8.1 Circuit with broadband directional couplers

Using the previously obtained blocks, a 2×2 passive optical multiplier circuit with fixed coefficients was implemented. Three output ports were analyzed in the experiment: one was considered the resulting multiplication channel, while the other two were used as control channels.

In the first part of the circuit, a combination of MZIs with phase shifts of 0 , $\pi/2$, and π and broadband directional couplers generates a fixed set of amplitudes and phases corresponding to the specified weights of the matrix elements. Next, after additional Y-splitters and MZIs with constructive and destructive interference, the required power distribution between the outputs is formed.

These interferometer variants, combined with Y-splitters and BDCs, were implemented in the KLayout design environment. The geometry of the corresponding structures is hereinafter referred to as **Circuit 6** and is shown in Figure 28.

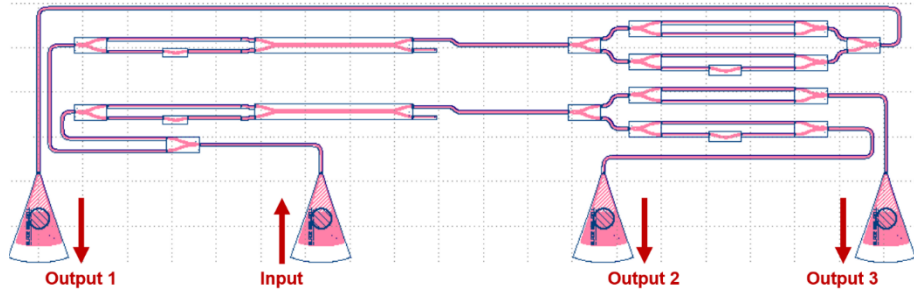


Fig. 28. General view of Circuit 6 in KLayout, consisting of grating couplers, y-splitters, waveguides, Mach-Zehnder interferometers and broadband directional couplers.

In the idealized lossless case, using the described Mach-Zehnder interferometers with phase shifts of 0 , $\pi/2$, and π , as well as the BDC, the expected transmittances at the three analyzed outputs of Circuit 6 are 0.125 , 0 , and 0.25 . These values were obtained from a model in which the Y-splitter divides the power equally between the arms and the BDC redistributes it among the outputs in accordance with the previously measured division ratios.

To verify the described circuit, numerical simulations were also performed in Lumerical INTERCONNECT; the system diagram is shown in Figure 29.

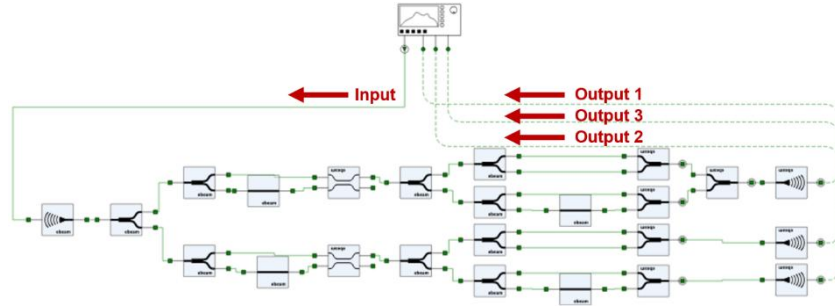


Fig. 29. Circuit 6, implemented in Lumerical INTERCONNECT, consisting of Optical network analyzer, grating couplers, y-splitters, waveguides and Broadband directional coupler.

Figure 30 shows the numerical and experimental spectra for Circuit 6.

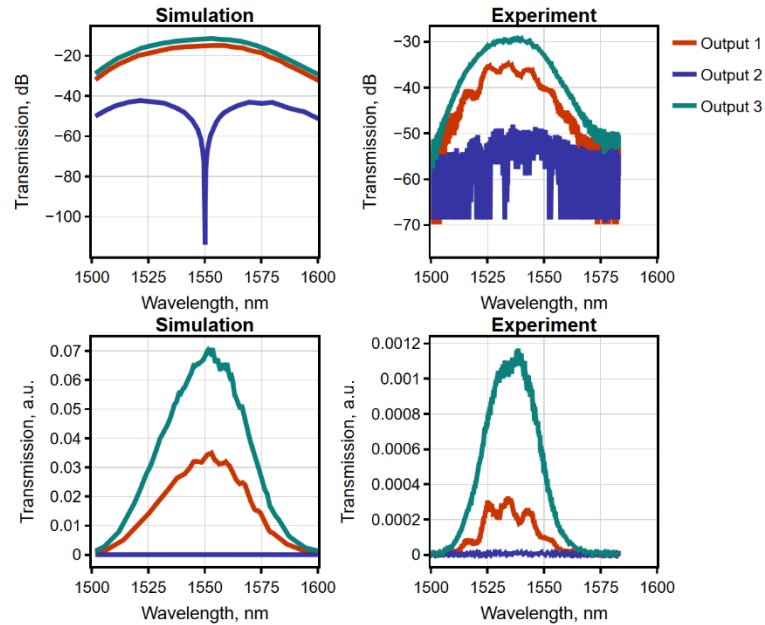


Fig. 30 below shows the numerical results and experimental data for Circuit 6.

Figure 31 shows the normalized and smoothed data.

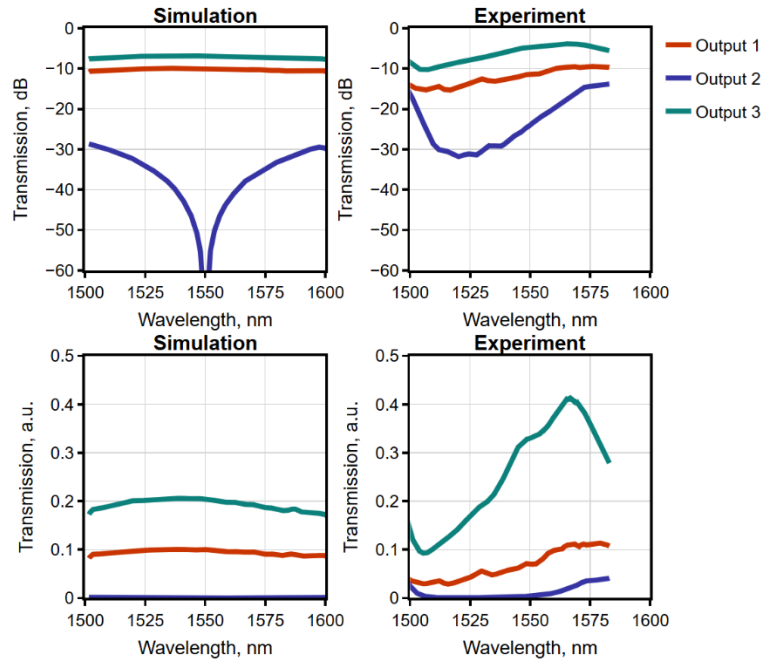


Fig. 31. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

Summary values at 1550 nm are given in Table 6.

Table 6. Output transmission values obtained for circuit 6.

		Output 1	Output 2	Output 3
Circuit 6	Theory	0.125	0	0.25
	Simulation	0.1	0	0.21
	Experiment	0.07	0	0.33
	Error	30%	0%	57%

Despite significant relative discrepancies for some individual outputs, the experimental spectra reproduce the shape of the numerical curves well, confirming the correctness of the circuit's functional implementation. High error values are primarily due to normalization to finite T_{num} and are particularly noticeable for channels with moderate power levels, where even small absolute shifts of 0.1–0.15 in transmission yield relative errors of 30–60%.

8.2 Circuit with multimode interferometers

In the second implementation of the 2×2 passive multiplier, the broadband BDCs were replaced with 2×2 multimode MMI interferometers. The configuration is built on the same principle: the first part of the circuit (MZI + MMI) specifies the amplitude and phase set, while the second part, with Y-splitters and MZIs, implements constructive and destructive interference and thus the coefficient matrix (Figure 32).

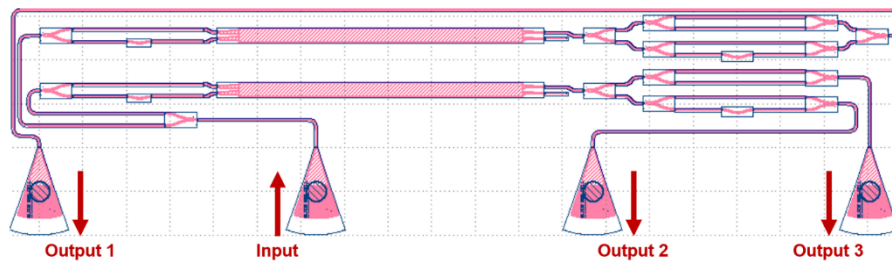


Fig. 32. General view of Circuit 7 in KLayout, consisting of grating couplers, y-splitters, waveguides, Mach-Zehnder interferometers and multimode interferometers.

In the idealized lossless case, the expected transmittances at the three analyzed outputs of Circuit 7 are 0.125, 0, and 0.25.

Figure 33 shows the numerical and experimental data for Circuit 7.

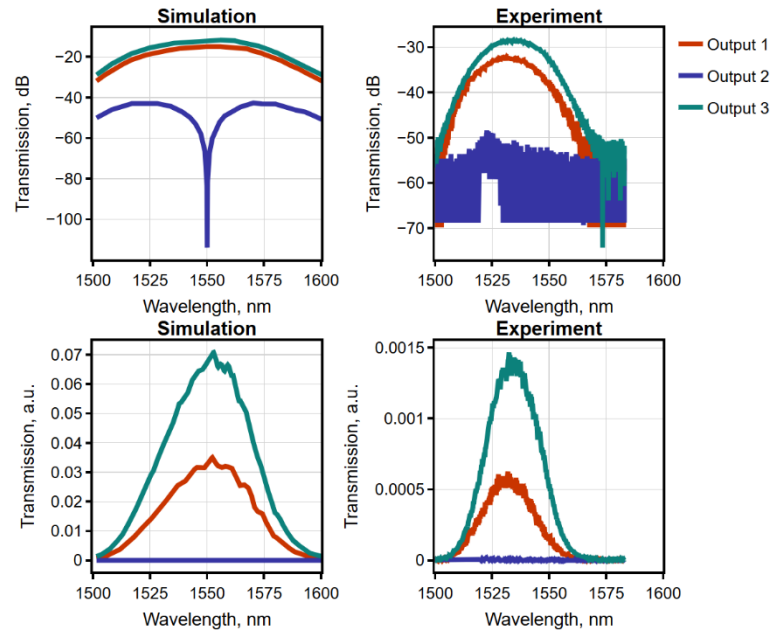


Fig. 33. Initial data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

Figure 34 shows the normalized and smoothed spectra.

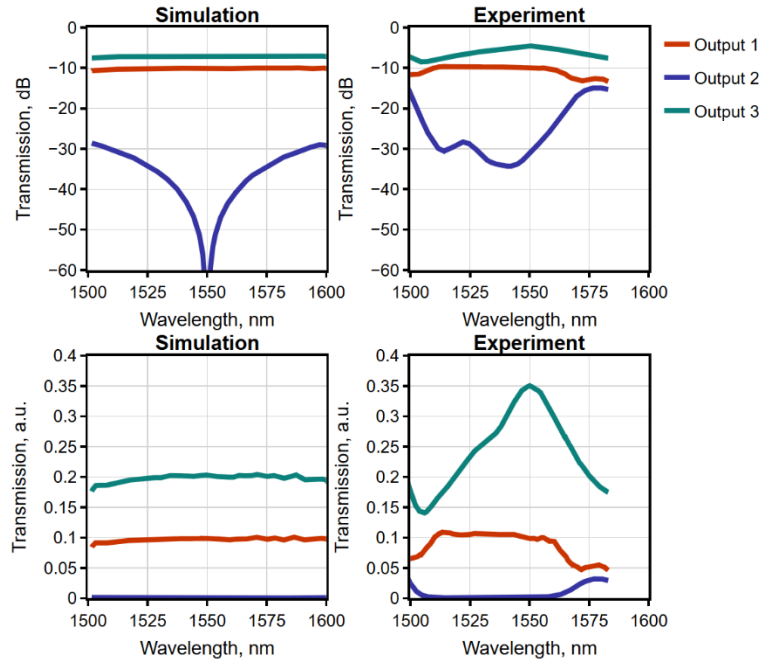


Fig. 34. Normalized and smoothed data obtained numerically and experimentally. The first row corresponds to the transmittance in dB, and the second to the transmittance in relative units. Color indicates correspondence to specific outputs on the diagram.

A statistical analysis of the impact of fabrication errors on the final output powers was also performed for the complete circuit using the same element parameters as in the previous calculations. The resulting distributions are shown in Figure 35.

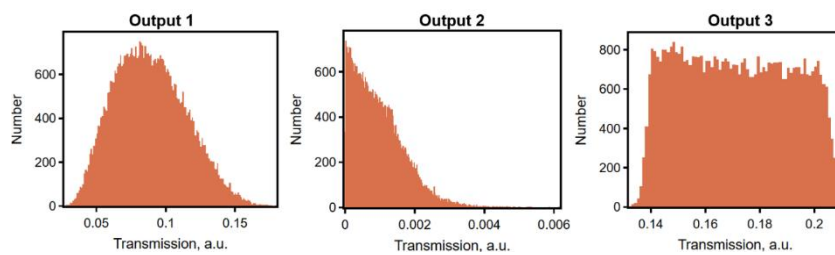


Fig. 35. Probability of power distribution at each of the outputs of Circuit 7.

The power values at outputs 1 and 2 are in excellent agreement with the experimental measurements. The value at output 3 deviates from the expected value. Unfortunately, without microscopy images of the circuit itself, it is difficult to determine the exact cause of this discrepancy. It is possible that fabrication inaccuracies introduced such errors into the system or that measurement errors occurred. However, it is also worth noting that the shapes of the experimental and numerical curves agree quite well. The summary values at a wavelength of 1550 nm are presented in Table 7.

Table 7. Output transmission values obtained for circuit 7.

		Output 1	Output 2	Output 3
Circuit 7	Theory	0.125	0	0.25
	Simulation	0.1	0	0.2
	Experiment	0.1	0	0.35
	Error	0%	0%	75%

Despite the magnitude of the relative error, the absolute power difference is approximately 0.1–0.15, and the spectral shape is fully reproducible. Therefore, this estimate is conservative and primarily reflects the metric's sensitivity to small T_{nm} values. As with the BDC variant, the spectral shape for all three outputs is in good agreement with the simulation, while the significant difference in power level at output 3 (0.20 versus 0.35) leads to a high relative error. Qualitatively, this means that the circuit implements the correct power distribution and suppression pattern. However, for use as an accurate matrix multiplier, additional phase calibration and consideration of the MMI and MZI process variations are required.

9. Conclusion

This work demonstrates a complete design, simulation, and experimental verification workflow for passive circuits on a silicon photonics platform using the SiEPIC PDK. Several classes of passive elements and their combinations were investigated. For Mach–Zehnder interferometers with various phase shifts between their arms, good quantitative agreement was achieved between the idealized lossless model, numerical simulations, and experiment. At a wavelength of 1550 nm, the relative errors for the two main outputs of Circuit 1 are approximately 2–9%, while the third output in destructive interference mode remains suppressed in both calculations and measurements.

Broadband directional couplers (BDCs) from the SiEPIC library exhibited pronounced power distribution asymmetry and high sensitivity to phase errors and process tolerances. When incorporating BDCs into more complex configurations, relative discrepancies between numerical simulation and experiment for individual outputs reached 20–50%, although the spectral shapes were still well reproduced.

Replacing BDCs with 2×2 multimode MMIs resulted in a more symmetrical and predictable power distribution. Approximating the MMI with a directional coupler with selected amplitudes and phases allowed reproduction of the ideal transmission coefficients with an accuracy of units and tens of percent, and experimental data showed a noticeable reduction in discrepancies for most channels compared to BDC-based configurations. The largest deviation was observed for one of the outputs, which can be attributed to systematic losses and the inaccuracy of the MMI approximation using an equivalent DC.

Overall, we can conclude that:

- The developed modeling and data processing methodology, including the use of foundry provided S-parameters, a test structure for normalization, and a unified spectral smoothing procedure, ensures reproducible comparison of theoretical, numerical, and experimental results.
- Mach–Zehnder interferometers and MMI based circuits demonstrate good quantitative agreement with calculations and can be considered passive circuit elements that are more tolerant of process variations compared with configurations based on broadband directional couplers.
- In the considered topologies, broadband directional couplers implement the correct power distribution and suppression pattern. However, their use as accurate matrix multiplication units requires additional phase calibration, explicit consideration of process variations, and, if necessary, refinement of the PDK models.

Thus, the conducted analysis demonstrates the practical applicability of the SiEPIC platform and the proposed data processing methodology for the design and debugging of more complex photonic integrated circuits, including programmable matrix multipliers based on variation tolerant combinations of MZI and MMI elements.

10. Summary

Table 8 summarizes the output transmittance values for all circuits.

Table 8. Summary of output transmittance values for all circuits

		Output 1	Output 2	Output 3	Average error
Circuit 1 (MZIs)	Theory	0.5	0.125	0	4%
	Simulation	0.46	0.11	0	
	Experiment	0.45	0.1	0	
	Error	2.2%	9.1%	0%	
Circuit 2 (BDCs)	Theory	0.125	0.2	0.28	37%
	Simulation	0.12	0.18	0.27	
	Experiment	0.07	0.14	0.4	
	Error	42%	22%	48%	
Circuit 3 (MMIs)	Theory	0.5	0.25	0.25	22%
	Simulation	0.43	0.22	0.22	
	Experiment	0.26	0.19	0.25	
	Error	40%	14%	12%	
Circuit 4 (MZIs + BDC)	Theory	0.25	0	0.5	31%
	Simulation	0.20	0	0.45	
	Experiment	0.13	0.06	0.33	
	Error	35%	0%	27%	
Circuit 5 (MZIs + MMIs)	Theory	0.25	0	0.5	10%
	Simulation	0.22	0	0.45	
	Experiment	0.21	0	0.56	
	Error	5%	0%	24%	
Circuit 6 (2x2 c BDCs)	Theory	0.125	0	0.25	29%
	Simulation	0.1	0	0.21	
	Experiment	0.07	0	0.33	
	Error	30%	0%	57%	
Circuit 7 (2x2 c MMIs)	Theory	0.125	0	0.25	25%
	Simulation	0.1	0	0.2	
	Experiment	0.1	0	0.35	
	Error	0%	0%	75%	

The average relative error is approximately 21%. On average, the error increases by 13% when using BDC instead of MMI on similar circuits.

11. General circuit

The circuit presented in this work, sent to SiEPIC production, is shown in Figure 36. The circuit size is 605 x 410 μm .

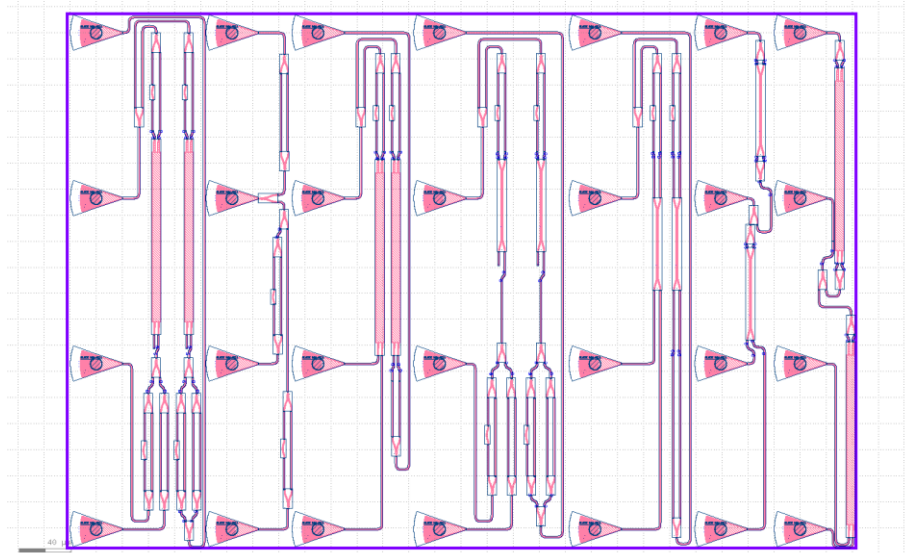


Fig. 36. The external appearance of the circuit designed in KLayout, which was sent to the SiEPIC fab.

General view of the chip produced at the SiEPIC factory together with other course participants (Figure 37).

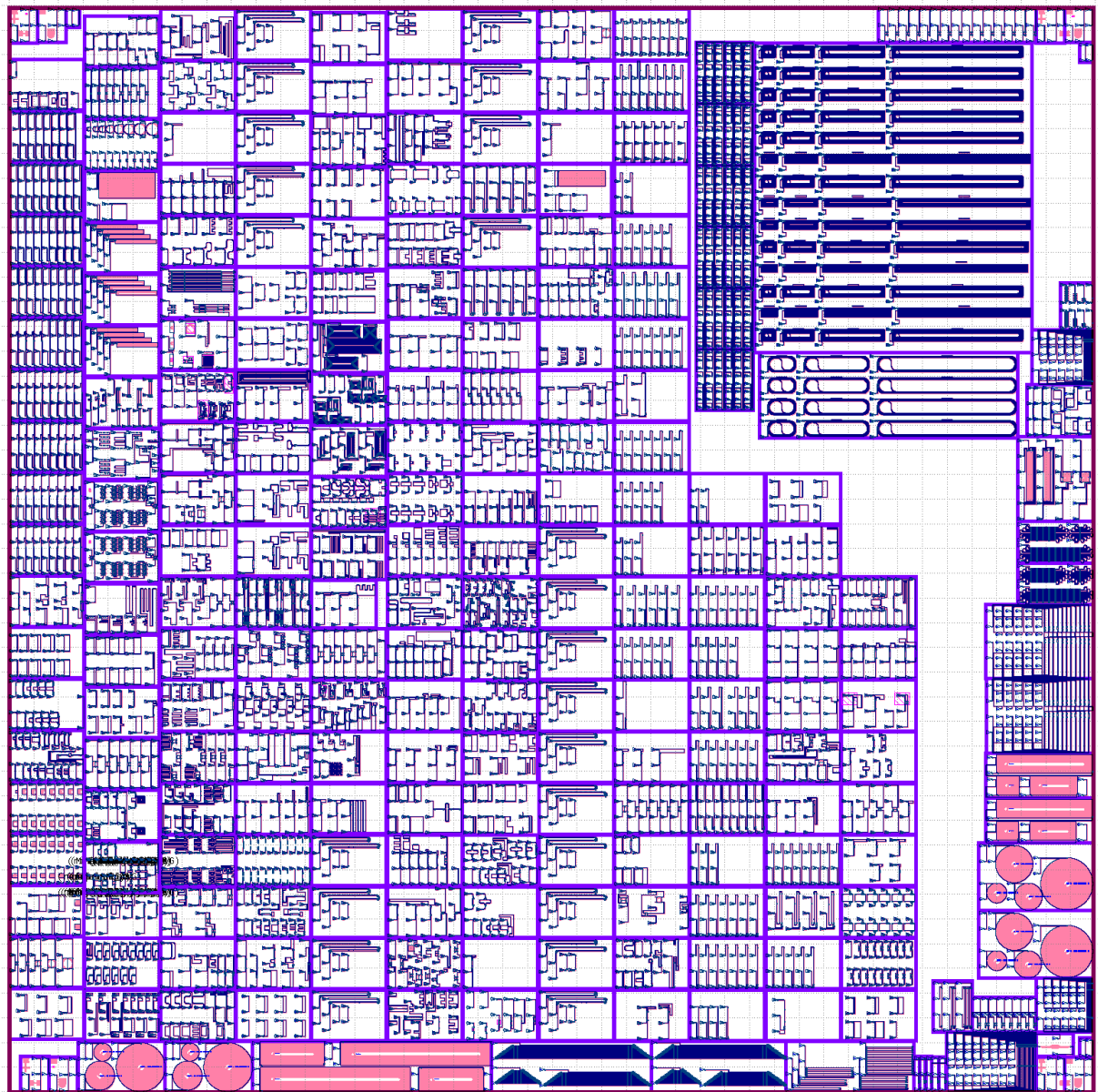


Fig. 37. External appearance of the circuit, which was produced at the SiEPIC factory.

12. Acknowledgment

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